NEBRASKA

Heat, Wealth, and Vegetation: Urban Heat Island Drivers in Lincoln, Nebraska

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Introduction

Motivation, Problem Statement, Research Question, Methods

Introduction, Motivation, Problem

- **<u>The Problem:</u>** Urban Heat Island (UHI) effects cause urban areas to be significantly hotter than rural surroundings due to impervious surfaces like concrete and asphalt.

- **<u>Environmental Inequity:</u>** Heat impacts are not evenly distributed, often exacerbating existing social disparities.

- **<u>The "Luxury Effect":</u>** Preliminary data suggests that wealthier neighborhoods in Lincoln benefit from more cooling infrastructure (trees/greenery) compared to low-income and renter-heavy areas.

- **<u>Objective:</u>** Investigate how landscape characteristics (vegetation/urban area) and socioeconomic variables (income/poverty) interact to influence heat exposure in Lincoln.

Research Question

- How do environmental and socioeconomic factors influence the spatial distribution of heat in Lincoln Nebraska?

Methods

- Exploratory Data Analysis & Visualization
- Correlation Analysis
- Heat Hot-Spot Identification
- Spatial Mapping & Clustering
- Machine Learning & Explainability

Exploratory Data Analysis & Visualization

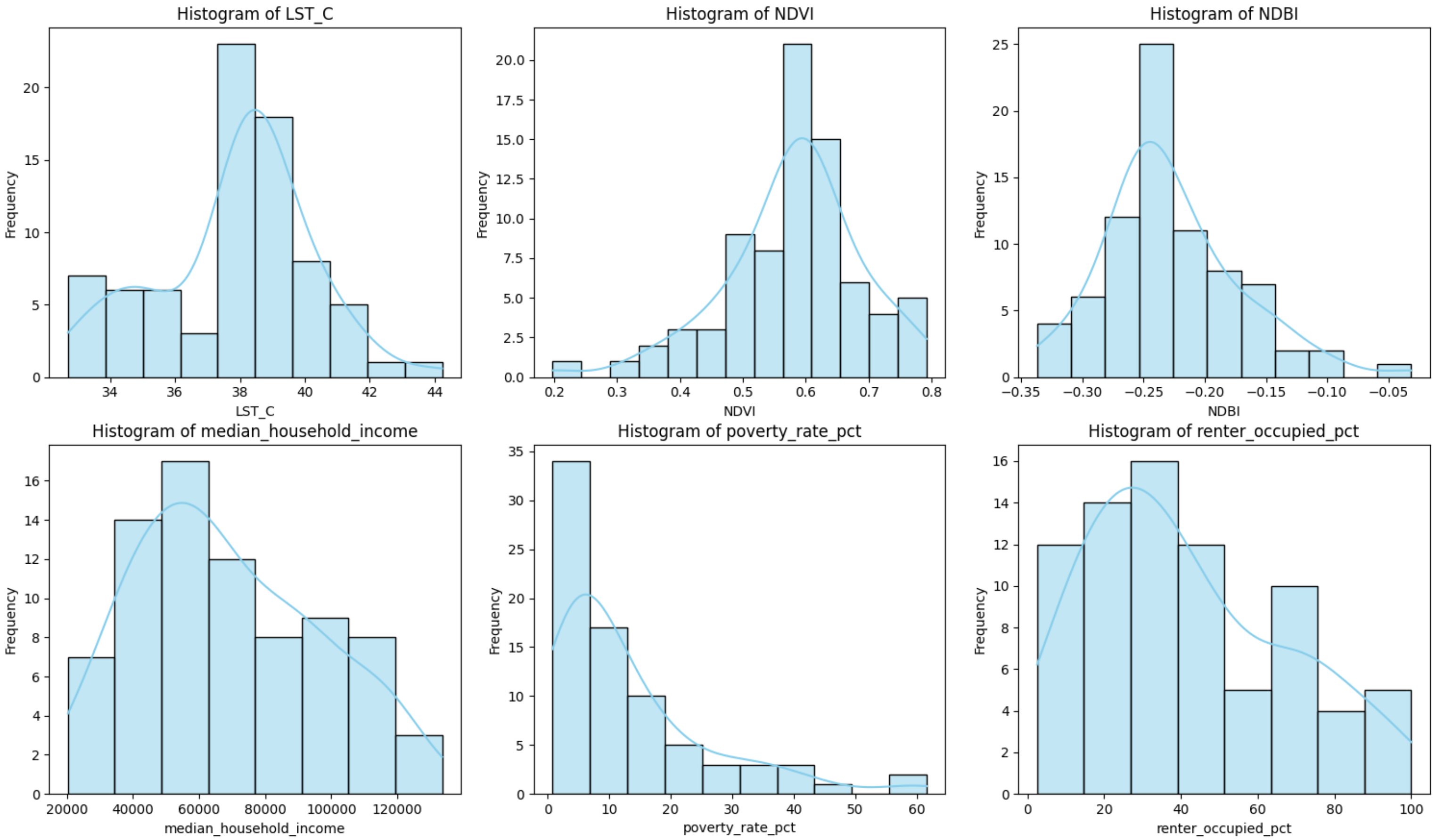
Preprocessing, EDA, Visualizing Bivariate Relationships

Preprocessing

- 1st Dataset: Landsat 8 & 9 Satellite imagery for land surface temperature
    - NDVI & NDBI data from satellite bands
- 2nd Dataset: Lincoln Census Tract boundary GEOJSON file
- 3rd Dataset: ACS Census socioeconomic variables

Zonal statistics were applied to aggregate raster-based satellite data to the census tract level. These outputs were then merged with ACS by standardizing GEOIDs and matching by year to create a master dataset

Exploratory Data Analysis



LST (Land Surface Temperature)

- Follows a normal distribution
- Most values cluster around 38-39 ℃

NDVI (Normalized Difference Vegetation Index)

- Left-skewed distribution
- Indicated most of Lincoln has a healthy, dense greenery

NDBI (Normalized Difference Built-up Index)

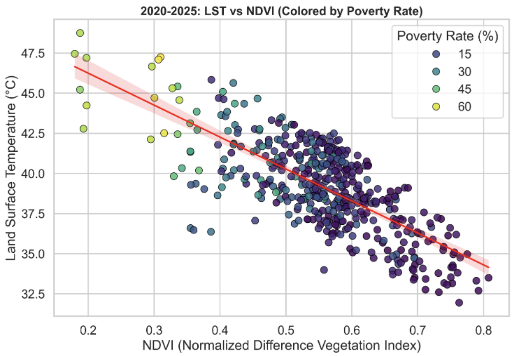
- Right-skewed distribution
- Shows a few areas with intense urban development
- Majority of Lincoln is less densely built

Socioeconomic Variables

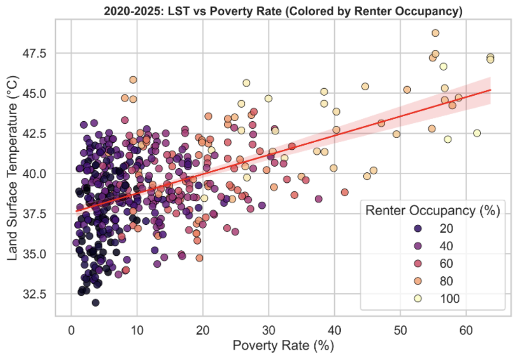
- All show right-skewed distributions
- Most neighborhoods have lower poverty and renter rates
- A small number of areas experience much higher economic challenges

Visualizing Bivariate Relationships

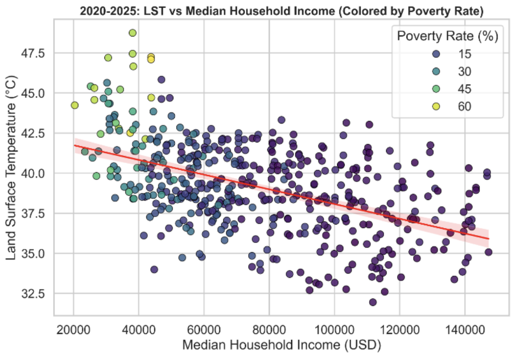
- **Linear Regression Fits:** Simplify complex data into a clear overall trend.
- **Confidence Intervals:** Shaded regions represent the statistical certainty of trends, ensuring reliable data
- **Variables:**
    - **LST:** Temperature of the earth – primary indicator of heat.
    - **NDVI:** Measures the vegetation. High values mean healthy vegetation; low values mean concrete/paved surfaces.
    - **Poverty Rate:** Percentage of population living below the federal poverty line.
    - **Median Income:** "Middle" income value of a household used as the indicator for neighborhood wealth.
    - **Renter Occupancy:** Proportion of residents who rent vs own.



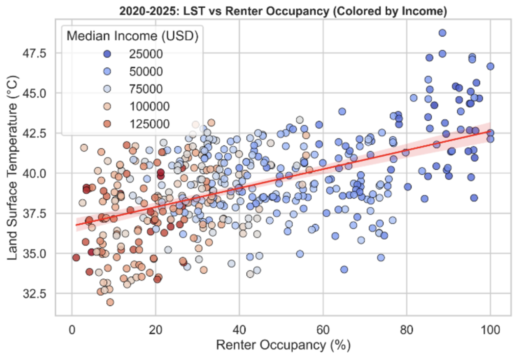
(a) Relationship between NDVI and land surface temperature (LST), colored by poverty rate.



(b) Relationship between poverty rate and LST, colored by renter occupancy.



(c) Relationship between median household income and LST, colored by poverty rate.



(d) Relationship between renter occupancy and LST, colored by median household income.

Heat Hot-Spot Identification

Welch's T-Test

Welch's t-test

$$t = \frac{\bar{X}_{hot} - \bar{X}_{cool}}{\sqrt{\frac{s_{hot}^2}{n_{hot}} + \frac{s_{cool}^2}{n_{cool}}}}$$

- Positive t - Increases with high LST
- Negative t - Increases with low LST
- $|t| > 2.1$ - Statistically significant

- Tells us the significance of a given variable in the context of the hottest and coolest tracts.
- Tells us if a given variable is more prominent in the hottest tracts or the coolest

t-values

Variable	t-value
Land Surface Temperature (C)	40.107
NDVI	-14.770
NDBI	13.103
Median Household Income	-13.620
Poverty Rate (%)	7.749
Renter Occupied (%)	37.788
65 Plus (%)	-7.097
65 Plus Count	-8.059
Elderly Living Alone (%)	3.000
Elderly Living Alone Count	-3.925
Disability (%)	2.832
Disability Count	-0.506
Nonwhite (%)	4.789
Nonwhite Count	4.510
Limited English Household (%)	1.210
Limited English Household Count	0.567

Most Significant Positive

- NDBI
- Poverty Rate
- Renter Occupied Percentage

Most Significant Negative

- NDVI
- Median Household Income
- 65 Plus

Correlation Analysis

Mathematics Utilized

- **Pearson correlation** (measures linear association):

$$r = \frac{\sum_{i=1}^{n}(LST_i - \overline{LST})(X_i - \bar{X})}{\sqrt{\sum_{i=1}^{n}(LST_i - \overline{LST})^2}\sqrt{\sum_{i=1}^{n}(X_i - \bar{X})^2}}$$

- **Spearman rank correlation** ρ (measures monotonic association using ranks R):

$$\rho = 1 - \frac{6\sum_{i=1}^{n}(R_{LST_i} - R_{X_i})^2}{n(n^2 - 1)}$$

- **Kendall's Tau** (measures monotonic association using concordant/discordant pairs):

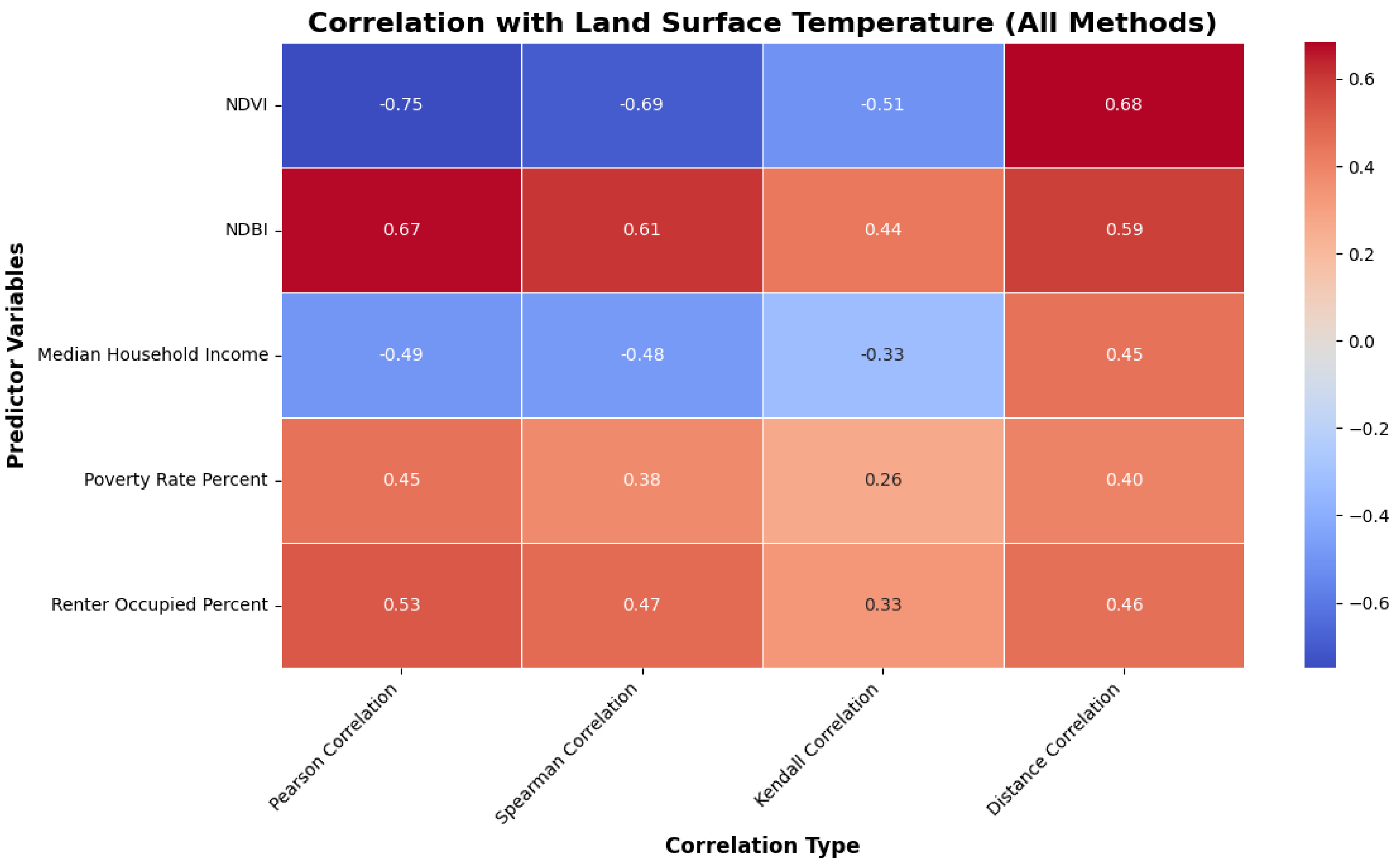
$$\tau = \frac{C - D}{\binom{n}{2}}$$

  - C = number of concordant pairs
  - D = number of discordant pairs
  - $\binom{n}{2} = \frac{n(n-1)}{2}$ is the total number of pairs

$$\tau = \frac{2(C - D)}{n(n-1)}$$

- **Distance Correlation** (measures any dependence, not just linear/monotonic and will always be nonnegative):

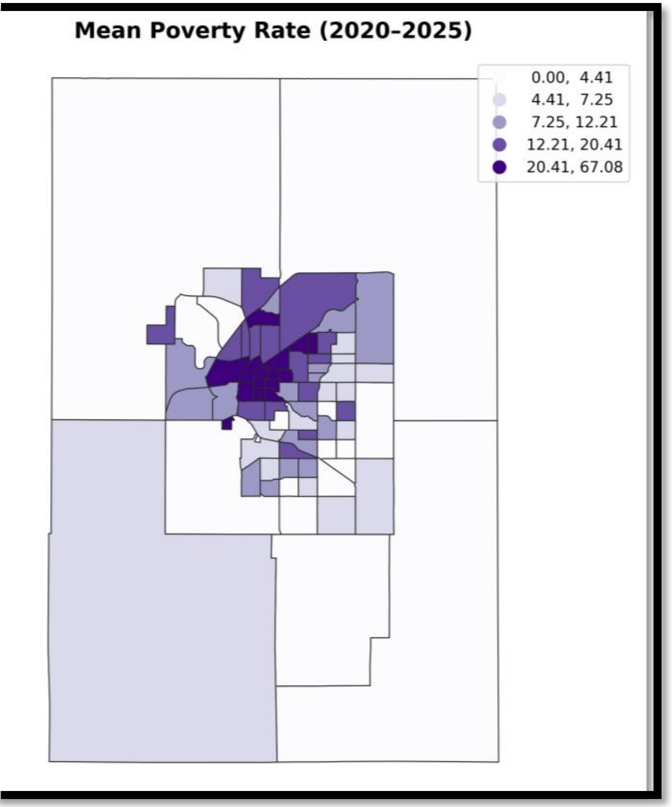
$$\mathrm{dCor}(X, Y) = \frac{\mathrm{dCov}(X, Y)}{\sqrt{\mathrm{dVar}(X)\,\mathrm{dVar}(Y)}}$$

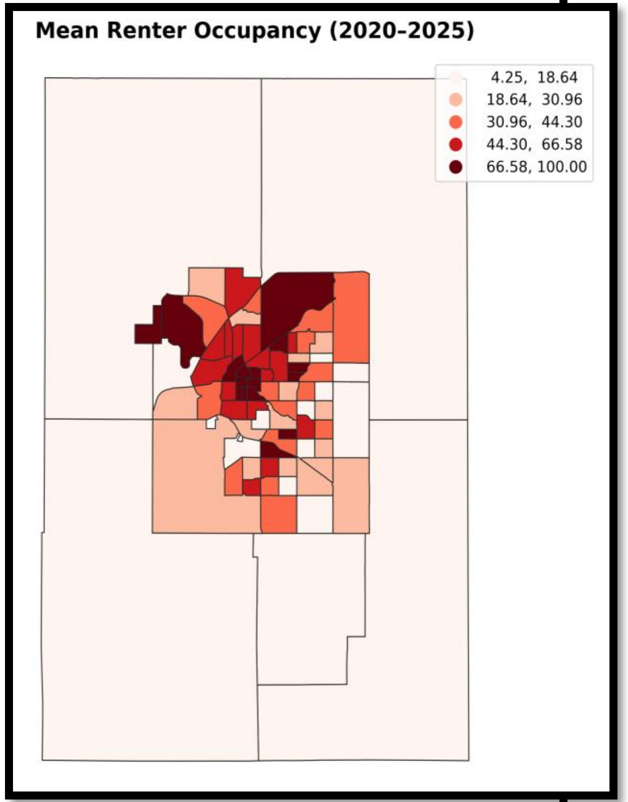

Correlation with Land Surface Temperature (All Methods)
Predictor Variables
NDVI
NDBI
Median Household Income
Poverty Rate Percent
Renter Occupied Percent
-0.75
-0.69
-0.51
0.68
0.67
0.61
0.44
0.59
-0.49
-0.48
-0.33
0.45
0.45
0.38
0.26
0.40
0.53
0.47
0.33
0.46
Pearson Correlation
Spearman Correlation
Kendall Correlation
Distance Correlation
Correlation Type
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

Spatial Mapping & Clustering

Why Spatial Mapping, Moran's I, Linking Heat & Inequality

Why Spatial Mapping?





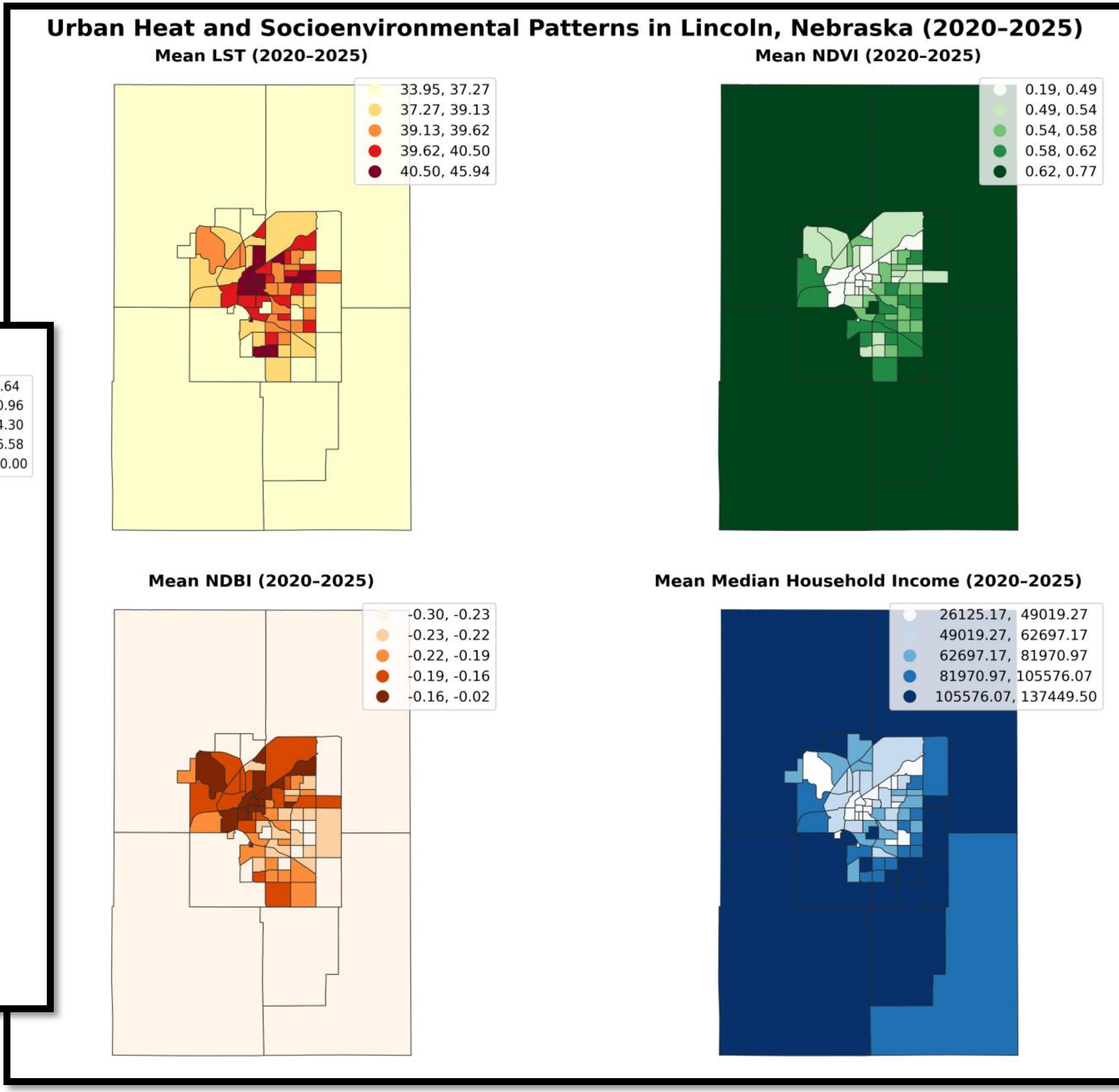


Figure 1: Variables across census tracts

Moran's I

- Global Moran's I = **0.528**
  - Strong, positive spatial autocorrelation
- LISA: High-High clusters -> Hotspots, Low-Low clusters -> cooler regions
- Heat is ***not randomly distributed***

$$I = \frac{n}{S_0} \cdot \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij}(LST_i - \overline{LST})(LST_j - \overline{LST})}{\sum_{i=1}^{n}(LST_i - \overline{LST})^2}$$

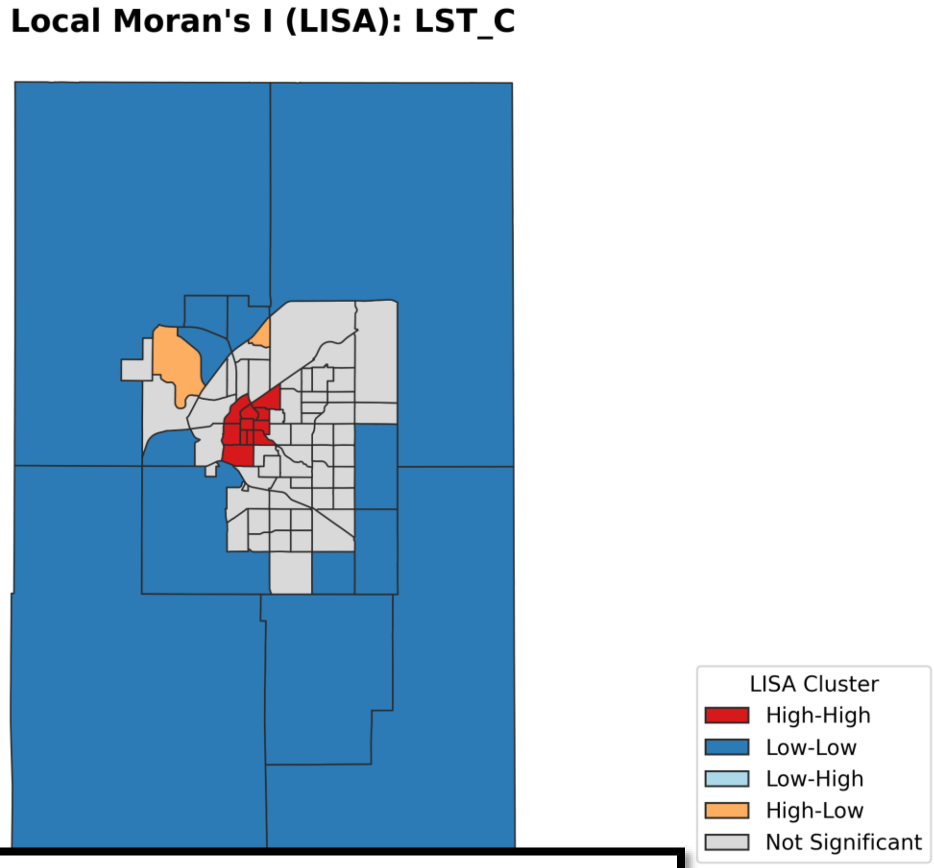


Table 4.6: LISA Cluster Distribution for LST

Cluster Type	Number of Tracts
High–High	11
Low–Low	13
High–Low	2
Low–High	0
Not Significant	55

Linking Heat & Inequality

Category	Number of Tracts
High LST – Low Income	17
High LST – High Income	3
Low LST – Low Income	3
Low LST – High Income	18

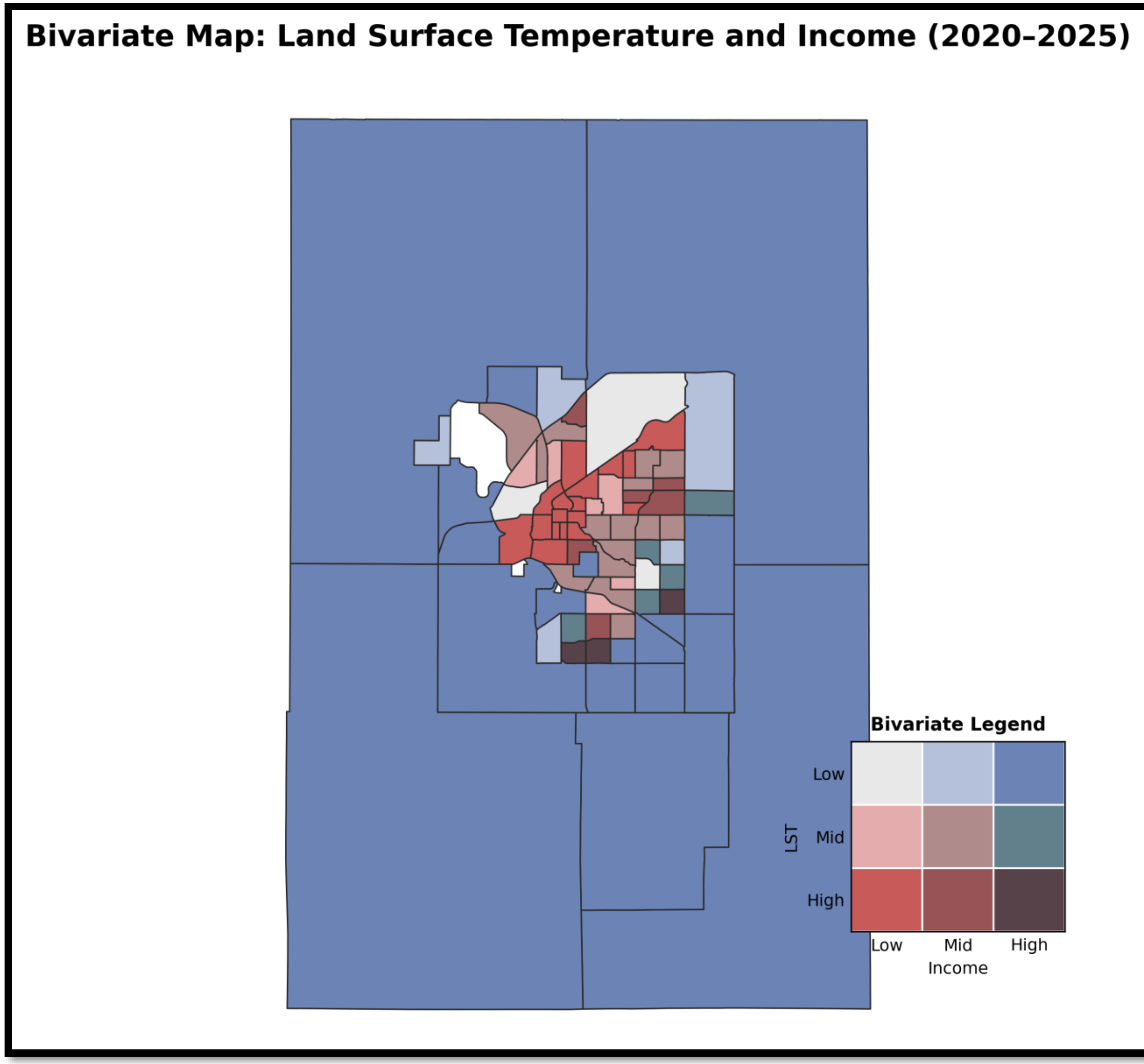


Figure 2: Bivariate map of land surface temperature and median household income

Linking Heat & Inequality

- Geographically Weighted Regression: allows relationships to vary across space
- NDVI: negative across all tracts -> Vegetation negative effect on LST.
- NDBI: ~ positive relationship with temperature -> buildings/concrete means higher temps.
- Poverty Pct: The effect of poverty on heat depends on the type of neighborhood
- Heat inequality ≠ social -> social + environment

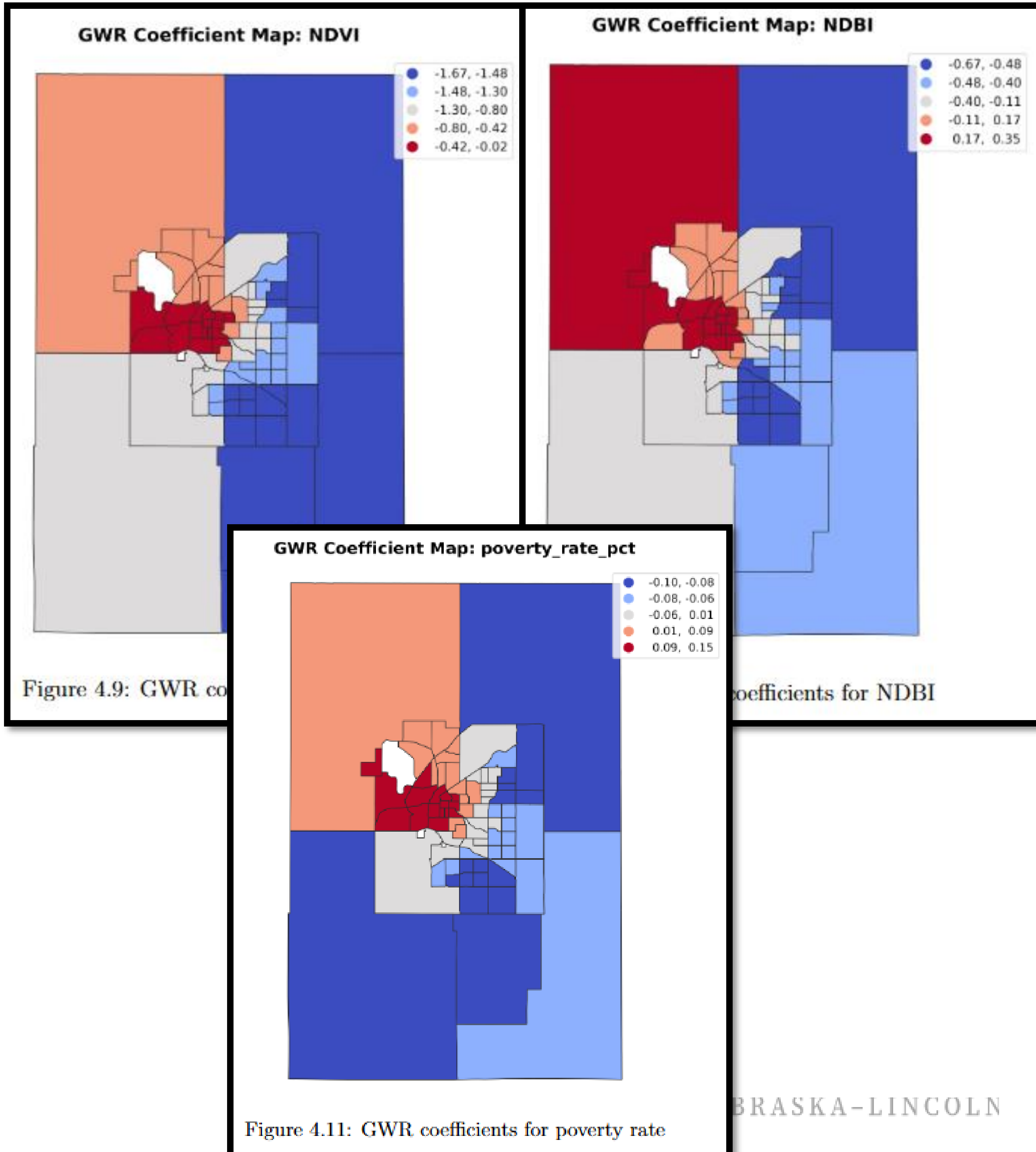


Figure 4.9: GWR co

coefficients for NDBI

Figure 4.11: GWR coefficients for poverty rate

Key Takeaway

- Urban heat is ***spatially clustered.***
- Strong link between
  - Heat exposure
  - Vegetation
  - Socioeconomic conditions
- Vulnerable communities face **higher heat risk**

Machine Learning & Explainability

Random Forest, SHAP Feature Attribution, Outputs & Analysis

Random Forest Model

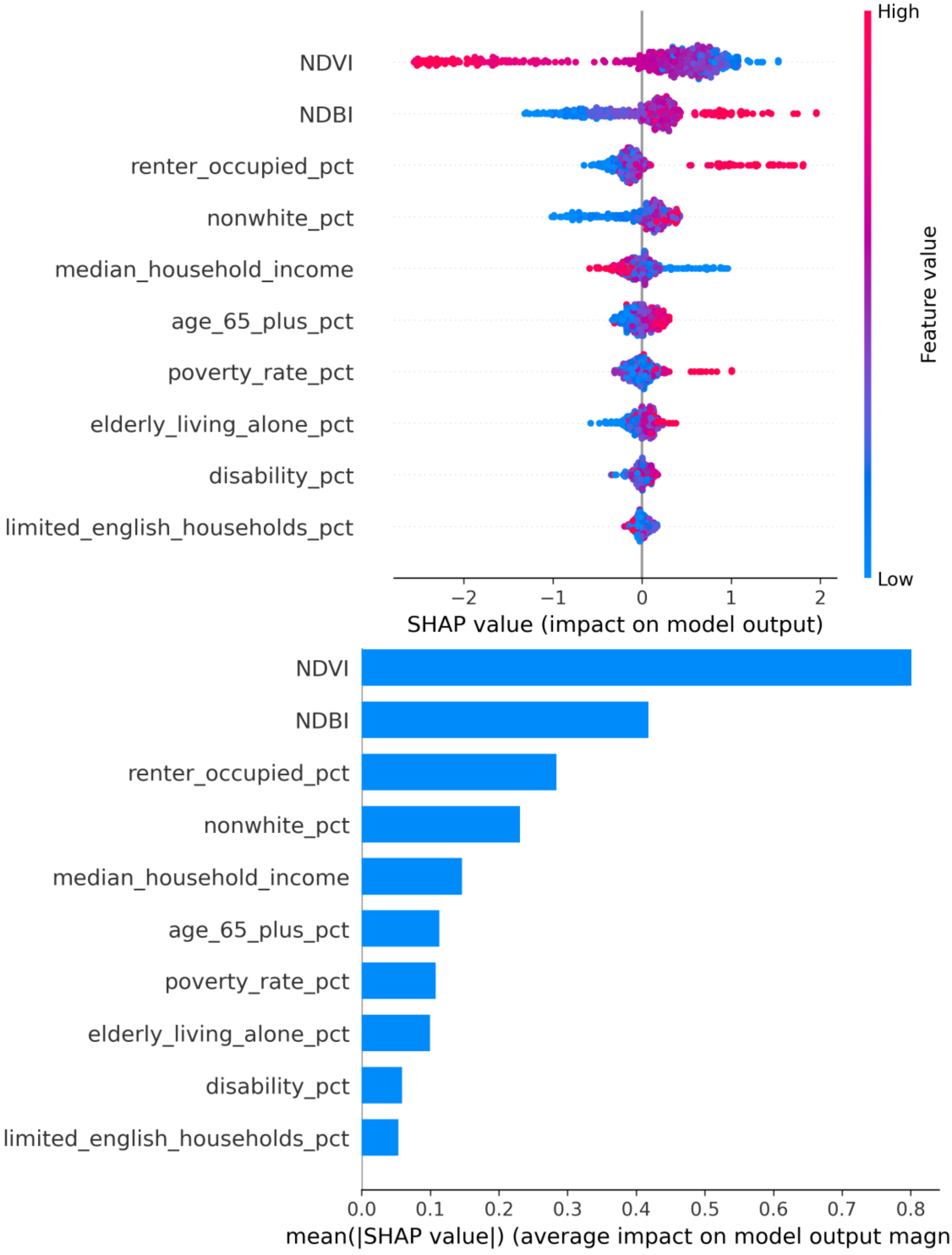
- Target Variable: Mean summer Land Surface Temperature (LST, °C) per census tract, derived from Landsat 8/9 satellite imagery via Google Earth Engine.
- Feature Matrix: 10 predictor variables across 464 tract-year observations (2020–2025). Two environmental, NDVI and NDBI, and eight ACS socioeconomic variables: median household income, poverty rate, renter occupancy, nonwhite %, age 65+, elderly living alone, disability %, and limited English households %.
- Model Specification: RandomForestRegressor with 800 trees, max_features="sqrt", min_samples_leaf=2. These settings balance variance reduction with generalization across the small census-tract dataset.
- Validation: Leave-One-Out Cross-Validation (LOOCV). Each of the 464 observations was held out once as a test point. This is the strictest possible CV for small datasets.

SHAP Feature Attribution

- Why SHAP: SHAP (SHapley Additive exPlanations) uses cooperative game theory to assign each feature a fair share of credit for each individual prediction, giving both the magnitude and direction of its effect on LST.
- Implementation: After fitting, shap.TreeExplainer was applied to the full feature matrix. TreeExplainer computes exact Shapley values for tree ensembles, producing one SHAP value per feature per year.
- Feature Summary Metrics: For each feature we computed:
  - (1) Mean |SHAP|, showing overall importance regardless of direction
  - (2) Mean signed SHAP, showing whether the feature tends to push LST up or down on average
  - (3) Correlation between raw feature value and its SHAP value, capturing the direction of the relationship .

Outputs and Analysis

- RMSE = 1.80°C, MAE = 1.51°C, R^2 = 0.57, meaning the model explains ~57% of LST variance across Lincoln tracts.
- SHAP values are highest for the environmental data (NDVI, NDBI), meaning they have more effect than the socioeconomic values
- The directional effect of each feature matches the previous findings, showing expected results



Conclusion

Discussion, Limitations & Implications, Future Work, References

Discussion

- **The "Luxury Effect" in Lincoln:** Confirms that heat exposure is not randomly distributed; it is deeply tied to the city's socioeconomic landscape.
- **Environmental Dominance:** NDVI and NDBI are the strongest predictors of local temperature. High-vegetation areas provide a critical cooling effect, while dense urban development traps heat.
- **Social Inequity:** Disadvantaged communities, specifically those with lower household incomes and higher poverty rates, experience significantly elevated heat exposure.
- **The Renter Gap:** One of the most striking findings is the main difference in housing tenurel hottest tracts average 91.9% renter occupancy, while coolest tracts average only 10.9%.

Limitations & Implications

- **Data Timeframe Restriction**
  - Analysis limited to 2020-2025 data
  - Lack of pre-COVID or older historical data restricts understanding of long term trends and pandemic impacts
- **Emerging Factors Not Fully Captured**
  - Did not analyze recent shifts in transportation, such as increased car and personal vehicle usage
- **Data Gaps and Missing Information**
  - Certain census tracts lacked income and demographic data due to presence of non-residential areas (e.g. prisons, airports)

Implications

- Results primarily reflect recent conditions, difficult to generalize to other time periods
- Highlight the importance of comprehensive data collection to capture complex urban dynamics
- Opportunity for future studies to integrate broader temporal and factor coverage for robust insights

Future Work

- **Extended Longitudinal Analysis:** Continuing to track these trends beyond 2025 to evaluate the long-term effectiveness of Lincoln’s 2021 Climate Action Plan.
- **Health Outcome Integration:** Future research could link this heat vulnerability map with specific public health data, such as heat-related hospitalizations or mortality rates in Lincoln.
- **Hyper-local Modeling:** Moving from census tracts to census block groups or even parcel-level data to provide even more granular insights for urban developers.
- **Community Outreach:** Developing targeted outreach programs for residents with limited English proficiency or those living alone, who may face more barriers to heat safety resources.

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Questions?

Thank you!

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Code: https://github.com/saharapandit/Math-In-The-City-Project/tree/main

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